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MASTER SYSTEM DIRECTORY (BP-01 TO BP-12)

HOW TO USE THIS BLUEPRINT AND LAYOUT BUNDLE

Every build plan, soil calculation, and planting matrix in this manual pairs directly with one of the twelve architectural blueprint plates (**BP-01 through BP-12**). All timber cuts use true lumber yard dimensions (1.5 by 9.25 in nominal 2x10 boards and 3.5 by 3.5 in nominal 4x4 posts), and all planting grids use true interior frame measurements.

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CRAFTSMAN REFERENCE STANDARDS

Lumber Standard: Nominal 2x10 (1.5 x 9.25 in actual), Nominal 4x4 (3.5 x 3.5 in actual), Nominal 2x4 (1.5 x 3.5 in actual).
Fasteners & Mesh: 3-in exterior star-drive (T25) structural wood screws, 1/2-in 19-gauge galvanized hardware cloth, and 3/4-in galvanized staples.

Section 1: System Orientation and the Square-Foot Matrix

COMPANION: BLUEPRINT PLATE 01 (BP-01)

Bio-Intensive Raised Bed Mechanics

Raised beds improve drainage, limit soil compaction, and help suppress weeds by lifting the root zone 9.25 inches (the true width of a nominal 2x10 board) above native grade. Because foot traffic stays strictly on the surrounding pathways, the growing medium stays loose and roots grow straight downward instead of fighting compacted clay. Intensive grid spacing produces higher harvest yields per square foot than traditional row planting, and raised soil drains and warms earlier in spring for earlier planting. Actual gains depend on crop cultivar, soil care, climate, and management.

The North-to-South Solar Hierarchy

Orient each bed with its long side running east to west, following the solar arc illustrated in **Blueprint Plate 01 (BP-01)**. Anchor tall trellised crops (pole beans, vining tomatoes, cucumbers) along the northern edge, place intermediate crops (peppers, eggplants, bush beans, brassicas) through the middle rows, and dedicate the sunny southern edge to low greens, alliums, and root crops. In the Northern Hemisphere, shadows fall northward, so this stepped arrangement prevents tall crops from shading shorter ones. (In the Southern Hemisphere, swap north and south.)

Planting Grids Use the Actual Interior

At the lumber yard, nominal 2x10 boards measure **1.5 inches thick by 9.25 inches tall**, and nominal 4x4 posts measure **3.5 by 3.5 inches**. Because wall thickness reduces the outside dimensions by 3.0 inches, your planting grid uses the true interior measurements: a **4x4 Classic Bed** has a 45.0 by 45.0 inch interior divided into sixteen cells of **11.25 by 11.25 inches**, and a **4x8 Workhorse Bed** has a 45.0 by 93.0 inch interior divided into thirty-two cells of **11.25 by 11.625 inches**. The four corner cells (and center cells in a braced 4x8 bed) lose a small notch to the 3.5 by 3.5 inch posts and recessed 2x4 cross-tie; the spacing tiers below are nominal, and actual spacing runs roughly 3 to 6 percent tighter, which is well within normal tolerance for intensive root zones.

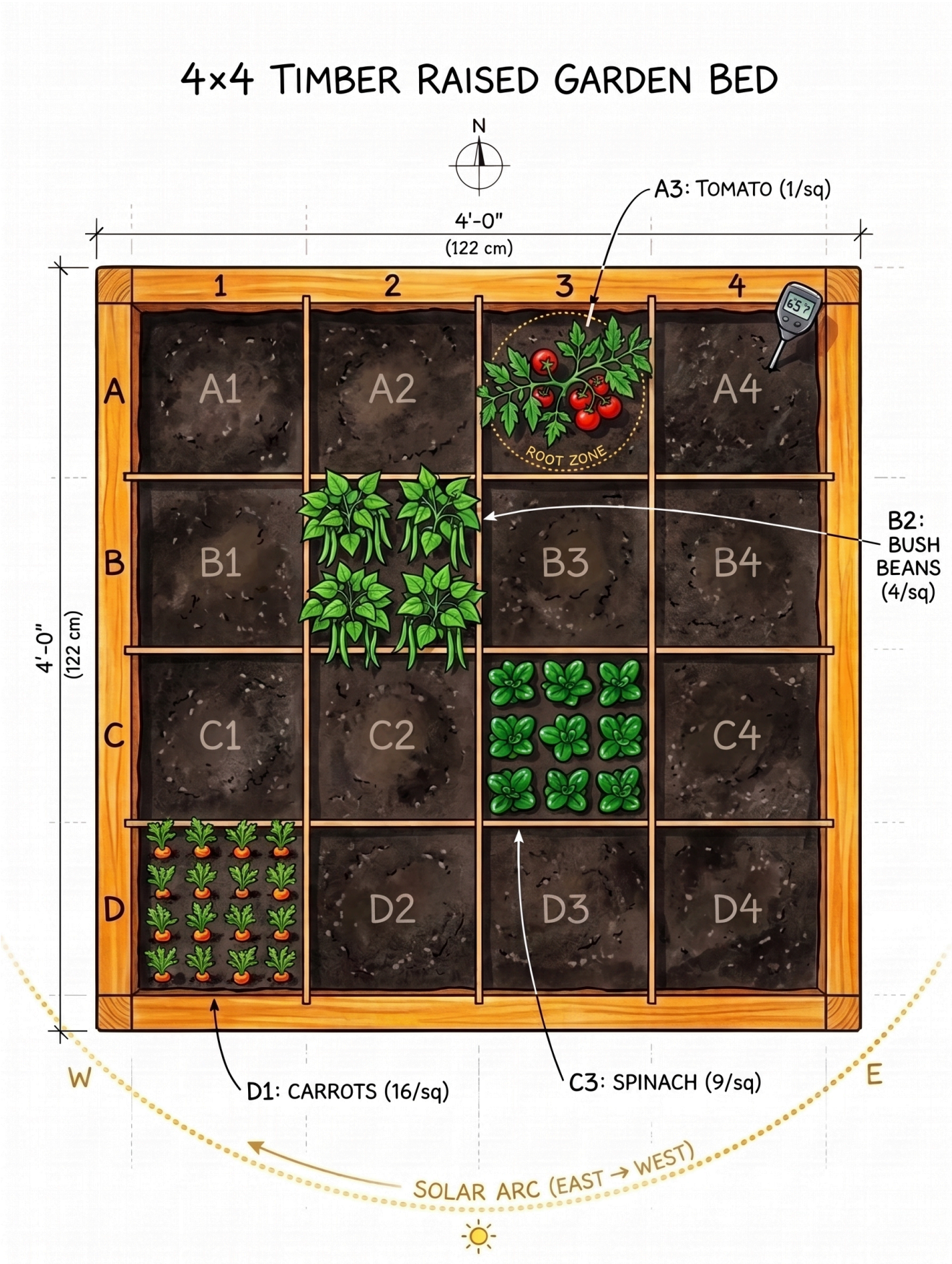
The Spacing Tiers

Every cell works as a discrete production unit, as mapped in **Blueprint Plate 01 (BP-01)**. Four core density tiers cover standard bed crops, and three trellised crops use dedicated linear counts:

Density Tier	Nominal Spacing	Grid Pattern	Approved Crop Roster
1 per square	12 in on center	Single center point	Tomatoes, sweet peppers, eggplants, cabbage, broccoli, kale, thyme, French marigold
4 per square	6 in on center	2 by 2 grid	Bush beans, butterhead and romaine lettuce, Genovese basil, Swiss chard, parsley, dill
9 per square	4 in on center	3 by 3 grid	Baby spinach, table beets, leeks, culinary chives, parsnips
16 per square	3 in on center	4 by 4 grid	Nantes and Danvers carrots, French breakfast radishes, scallions

TRELLIS SPECIALS AND ERGONOMIC REACH RULE

Trellis specials (grown on twine or a vertical trellis): sugar snap peas at 8 per square, pole beans at 4 per square, and slicing cucumbers at 2 per square.
Reach rule: every cell should sit within about 24 inches of an outer edge you can stand beside. All beds in this bundle are designed around that limit.


4×4 TIMBER RAISED GARDEN BED
N
4'-0"
(122 cm)
A3: TOMATO (1/sq)
1
2
3
4
A
B
C
D
A1
A2
A4
B1
B3
B4
C1
C2
C4
D2
D3
D4
ROOT ZONE
65°F
B2: BUSH BEANS (4/sq)
4'-0"
(122 cm)
W
E
D1: CARROTS (16/sq)
C3: SPINACH (9/sq)
SOLAR ARC (EAST → WEST)

Section 2: Build Plan 01, The 4x4 Classic Bed

COMPANION: BLUEPRINT PLATE 02 (BP-02)

The 4x4 Classic Bed has a 16.0 square foot outside footprint and a 45.0 by 45.0 inch interior frame (14.06 sq ft gross; 13.72 sq ft net planting surface after subtracting the four 3.5 by 3.5 inch interior corner posts). Two opposite boards run the full 48.0 inches, while the other two boards are cut to 45.0 inches so they nest cleanly between the outer walls (48.0 minus 1.5 minus 1.5 equals 45.0 in), as illustrated in the **Blueprint Plate 02 (BP-02)** exploded corner schematic. A 48.0 inch span requires no mid-span brace. Build with rot-resistant untreated cedar or redwood, or modern ground-contact pressure-treated lumber (ACQ or CA-C), using exterior-rated structural screws.

OUTSIDE SIZE	INTERIOR FRAME (NET)	GROSS VOLUME (NET SOIL)	OUTSIDE DIAGONAL CHECK
48.0 x 48.0 x 9.25 in	**45.0 x 45.0 in (14.1 / 13.7 sq ft)**	**10.8 cu ft / 0.40 cu yd (10.6 net)**	**67.9 in (67-7/8 in on tape)**

Verified Lumber Yard Buy List and Shop Cut List

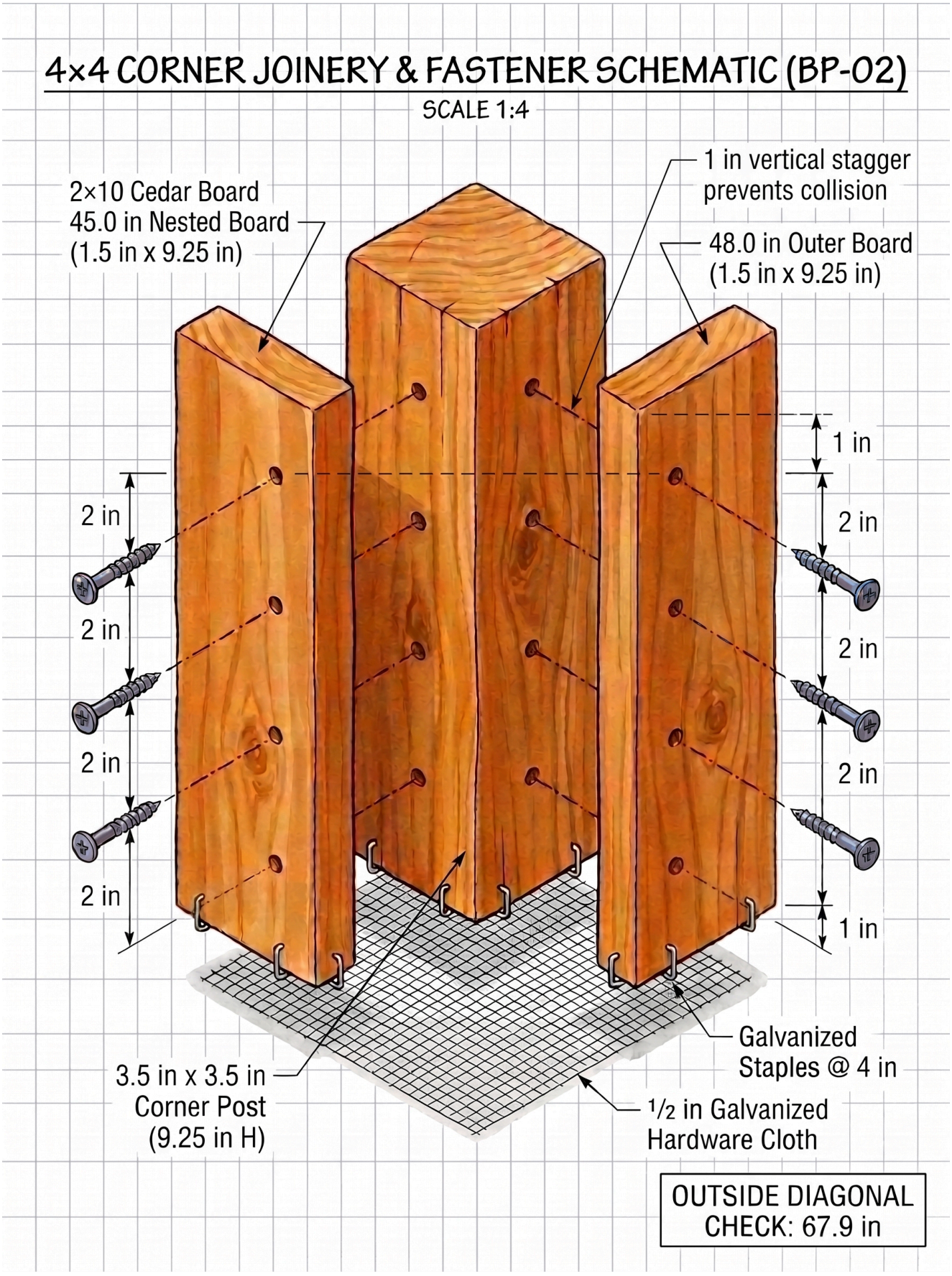
Part / Material	Exact Cut Size	Lumber Yard Buy List and Shop Notes
2 Long Outer Walls	2 pcs @ 48.0 in	**Buy two 8-ft (96 in) 2x10 boards.** Cut one 48.0 in outer board and one 45.0 in nested board from each 96 in plank (93.0 in used, leaving 3.0 in spare for saw kerf).
2 Nested Side Walls	2 pcs @ 45.0 in	
4 Corner Posts	4 pcs @ 9.25 in	**Buy one 4-ft (48 in) 4x4 post.** Four 9.25 in corner posts use 37.0 in total, leaving 11.0 in spare.
Structural Screws	32 pcs @ 3 in	Exterior star-drive (T25) structural wood or deck screws (8 per corner post).
Rodent Mesh and Staples	48 x 48 in mesh	1/2-in, 19-gauge galvanized hardware cloth plus about 48 galvanized 3/4-in staples.

Five-Step Assembly Sequence (See Blueprint Plate 02)

1. **Square the timber:** Cross-cut every board with a carpenter's speed square so all ends meet at a true 90 degree angle. Lightly sand top edges with 80-grit sandpaper.
2. **Mark and pre-drill staggered screw heights:** As shown in **Blueprint Plate 02 (BP-02)**, each corner post takes four screws through each of two adjoining faces. On the two **48.0 in outer boards**, mark screw heights at **1, 3, 5, and 7 in** from the bottom edge (1 in from bottom/top, 2 in between screws). On the two **45.0 in nested boards**, mark screw heights at **2, 4, 6, and 8 in** from the bottom edge. This 1-inch vertical stagger prevents 3-inch screws from colliding inside the post. Drill 1/8-in pilot holes.
3. **Attach the long walls:** Lay a 48.0 in board flat with its inside face up. Set a 9.25 in post at each end with its outer face inset **1.5 in** from the board tip and its top flush with the board top. Drive four 3-in screws per post. Repeat with the second 48.0 in board.
4. **Join the short walls and square the frame:** Stand the two 48.0 in assemblies facing each other, fit the 45.0 in nested boards into the 1.5 in end pockets against the posts with bottom edges flush, and drive four screws per post face. Measure outside corner to outside corner across both diagonals: each diagonal must read **67.9 in (67-7/8 in)** within 1/8 in. Nudge the frame until both match.
5. **Install the rodent barrier:** Turn the frame upside down, unroll the 1/2-in hardware cloth across the bottom, and drive a 3/4-in galvanized staple every 4 inches into the board edges and posts. Wear leather gloves and fold sharp wire ends flat before setting the bed on level ground.

4×4 CORNER JOINERY & FASTENER SCHEMATIC (BP-02)

SCALE 1:4



Section 3: Build Plan 02, The 4x8 Backyard Workhorse

COMPANION: BLUEPRINT PLATE 03 (BP-03)

A 4x8 bed has a 45.0 by 93.0 inch interior frame (29.06 sq ft gross; 28.72 sq ft net open surface at the rim) and holds 22.40 cubic feet of soil to the rim (22.07 cu ft net after subtracting the four corner posts and recessed cross-tie). Depending on moisture, that soil weighs **700 to 1,200 pounds**, and lateral soil load can bow unbraced 96.0 inch boards outward at mid-span. As detailed in **Blueprint Plate 03 (BP-03)**, this design adds one 45.0 inch 2x4 cross-tie flat across the center, recessed 4.0 inches below the top rim (leaving 4.0 in of clear planting depth above the 1.5 in thick tie and 3.75 in below it). **Shop rule:** any wall longer than 72 inches gets a mid-span tie along the boundary between columns 4 and 5.

OUTSIDE SIZE	INTERIOR FRAME (NET)	GROSS VOLUME (NET SOIL)	OUTSIDE DIAGONAL CHECK
48.0 x 96.0 x 9.25 in	**45.0 x 93.0 in (29.1 / 28.7 sq ft)**	**22.4 cu ft / 0.83 cu yd (22.1 net)**	**107.3 in (107-5/16 in on tape)**

Verified Lumber Yard Buy List and Shop Cut List

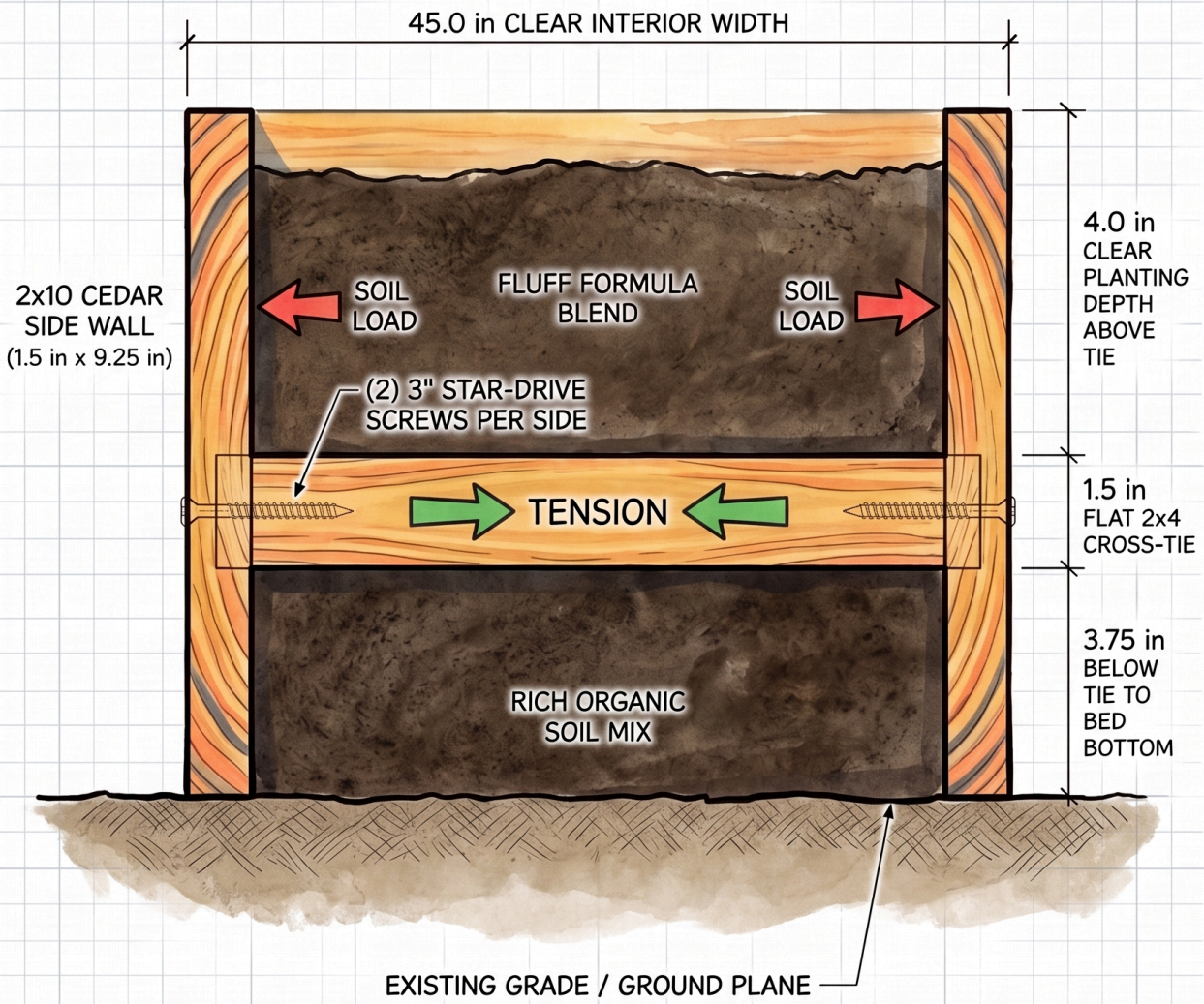
Part / Material	Exact Cut Size	Lumber Yard Buy List and Shop Notes
2 Long Side Walls	2 pcs @ 96.0 in	**Buy three 8-ft 2x10 boards total.** Trim two side boards square to 96.0 in if rough mill ends run long.
2 Nested End Walls	2 pcs @ 45.0 in	Cut both 45.0 in nested end boards from the third 96 in 2x10 board (90.0 in used, 6.0 in spare for kerf).
4 Corner Posts	4 pcs @ 9.25 in	**Buy one 4-ft 4x4 post.** Cut four 9.25 in corner posts flush with top and bottom board edges.
1 Center Cross-Tie	1 pc @ 45.0 in	**Buy one 8-ft (or 4-ft) 2x4 stud.** Installed flat at mid-span, recessed 4.0 in below the top rim.
Screws and Hardware Cloth	36 screws + mesh	36 exterior 3-in star-drive screws (32 at corners, 4 at cross-tie) + 48 x 96 in 1/2-in mesh + ~72 staples.

Five-Step Assembly Sequence (See Blueprint Plate 03)

1. **Lay out and square:** Place the two 96.0 in side boards and two 45.0 in end boards on level ground as a 48 by 96 in outline and check each corner with a carpenter's framing square.
2. **Build the corners:** Use the Section 2 staggered screw pattern: mark screws at **1, 3, 5, and 7 in** from the bottom on the 96.0 in side boards, and at **2, 4, 6, and 8 in** on the 45.0 in end boards. Attach a post inset 1.5 in from each end of both side boards, stand them up, and fit the 45.0 in end boards between them. Drive four 3-in screws per post face (32 corner screws in all).
3. **Square the frame:** Measure both outside diagonals corner to corner; each diagonal must read **107.3 in (107-5/16 in)** within 1/8 in.
4. **Install the recessed mid-span tie:** As shown in **Blueprint Plate 03 (BP-03)**, mark the center of each side board (48.0 in from each end). Mark a horizontal line **4.0 in below the top rim**; the top face of the flat 2x4 cross-tie (1.5 in thick by 3.5 in wide) sits on that line, leaving 4.0 in above the tie and 3.75 in below the tie to the bed bottom (4.0 + 1.5 + 3.75 = 9.25 in). Clamp the tie tight, drill pilot holes, and drive two 3-in star-drive screws through each side wall into the end grain of the tie (4 screws total) to lock the center interior width at **45.0 in**.
5. **Install the rodent barrier:** Turn the frame over and staple the 48 by 96 in hardware cloth every 4 inches into the bottom board edges and posts.

4×8 CROSS-TIE DETAIL (BP-03)

SCALE 1:8 TRANSVERSE CROSS-SECTION



FULL BED DIAGONAL: 107.3 in

Section 4A: Build Plan 03, The 2x8 Walkway Runner

COMPANION: BLUEPRINT PLATE 04 (BP-04)

Narrow zones along boundary fences, side yards, or house foundations make reaching across a 4-foot bed difficult from one side. As illustrated on the left half of **Blueprint Plate 04 (BP-04)**, the 2x8 Walkway Runner is built **24.0 inches wide outside** (21.0 by 93.0 in interior frame), so every cell sits well within a comfortable 24-inch single-side pathway reach. Build it using the Section 2 staggered screw pattern and the Section 3 mid-span tie method.

OUTSIDE SIZE	INTERIOR FRAME (NET)	GROSS VOLUME (NET SOIL)	SQUARING DIAGONAL
24.0 x 96.0 x 9.25 in	**21.0 x 93.0 in (13.6 / 13.2 sq ft)**	**10.5 cu ft / 0.39 cu yd (10.2 net)**	**99.0 in (98.95 in exact)**

Verified Lumber Yard Buy List and Shop Cut List

Part / Material	Exact Cut Size	Lumber Yard Buy List and Shop Notes
2 Long Side Rails	2 pcs @ 96.0 in	**Buy three 8-ft 2x10 boards total.** Two full 96.0 in boards form the front and rear walls.
2 Nested End Boards	2 pcs @ 21.0 in	Cut both 21.0 in nested end boards from the third 2x10 board (21.0 + 1.5 + 1.5 = 24.0 in outside width).
4 Corner Posts	4 pcs @ 9.25 in	**Buy one 4-ft 4x4 post.** Yields all four 9.25 in corner posts with 11.0 in spare.
1 Mid-Span Cross-Tie	1 pc @ 21.0 in	**Buy one 2x4 stud.** Cut to 21.0 in and install flat 4.0 in below the rim at the 48.0 in center mark.
Screws and Hardware Cloth	36 screws + mesh	36 exterior 3-in star-drive screws + one 24 x 96 in piece of 1/2-in hardware cloth + ~60 staples.

Five-Step Assembly Sequence (See Blueprint Plate 04)

1. **Cut end boards and tie to 21.0 inches:** Confirm that both 2x10 end boards and the 2x4 mid-span cross-tie measure an identical **21.0 in** so interior width stays uniform from end to end.
2. **Pre-drill staggered corner holes:** Mark **1, 3, 5, and 7 in** from the bottom on the 96.0 in side rails, and **2, 4, 6, and 8 in** on the 21.0 in end boards. Inset posts 1.5 in from the ends of both 96.0 in rails and drive four 3-in screws per post.
3. **Join ends and check the 99.0-inch diagonal:** Fasten the 21.0 in end boards between the long rails. Measure outside corner to outside corner in both directions until both diagonals read **99.0 in (98-15/16 in)**.
4. **Install the 21.0-inch mid-span cross-tie:** Mark the 48.0 in center point on both long rails, measure 4.0 in down from the top rim, and fasten the flat 21.0 in 2x4 cross-tie with two 3-in screws through each side rail.
5. **Attach mesh and site along your fence:** Staple the 24 by 96 in hardware cloth every 4 inches along the bottom rim. When placing the runner against a boundary fence, leave a **2 to 3 inch air gap** behind the rear board for airflow.

CRAFTSMAN ERGONOMIC NOTE: WHY 21.0 INCHES INSIDE WORKS

Dividing the 21.0 by 93.0 in interior into 2 rows of 8 columns yields sixteen cells measuring **10.5 in deep by 11.625 in wide**. As shown in the **Blueprint Plate 04** reach arc, every square sits within a 24.0 in reach from the front walkway.

Section 4B: Build Plan 04, The 6x6 Corner L-Bed

COMPANION: BLUEPRINT PLATE 04 (BP-04)

Detailed on the right half of **Blueprint Plate 04 (BP-04)**, the 6x6 Corner L-Bed wraps a 90 degree boundary corner with two 24.0 inch wide arms running 72.0 inches along the outer edges and framing a walk-in **48.0 by 48.0 inch inner courtyard** (20.0 sq ft footprint, 17.06 sq ft gross interior area equal to 2,457.0 sq in, and 16.64 sq ft net of the five corner posts). Because no wall exceeds 72.0 inches, no mid-span cross-tie is needed.

6X6 CORNER L-BED CUT LIST AND BUY LIST (BLUEPRINT PLATE 04 GEOMETRY)

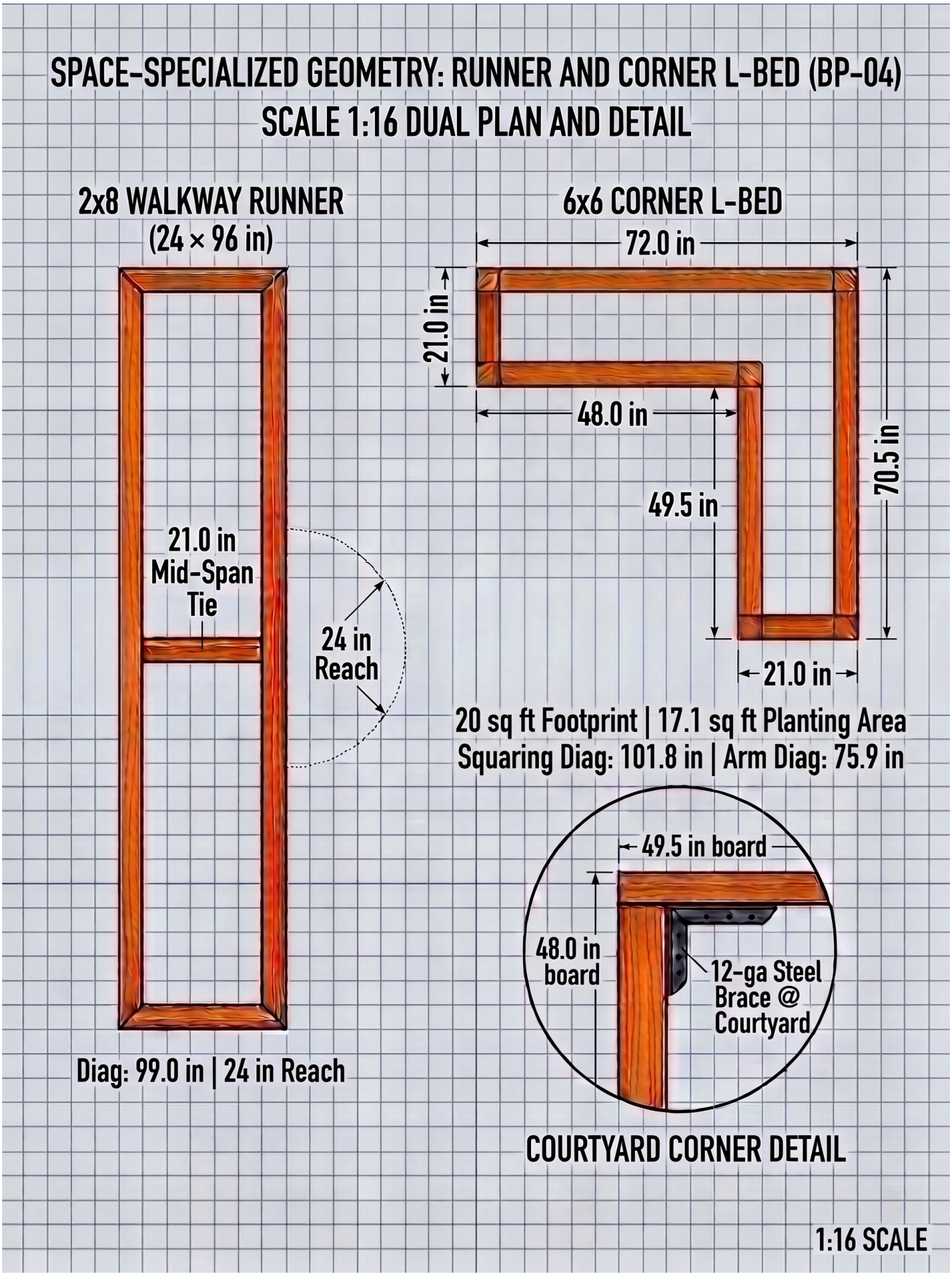
Buy List (Including 5th Post): Four 8-ft 2x10 boards (cutting 72.0 + 21.0 in, 70.5 + 21.0 in, 49.5 in, and 48.0 in) + **one 6-ft (or 8-ft) 4x4 post** (five 9.25 in posts require 46.25 in of length plus saw kerfs, which exceeds a single 4-ft post) + 43 exterior 3-in screws + one 12-gauge galvanized 90 degree steel corner brace + six 1-1/4 in structural screws + two 24-in wide hardware cloth strips (72 in and 50 in, overlapped 2 in) + ~72 staples.
Six 2x10 Board Cuts: (1) **Top/Left Outer Wall:** 72.0 in. (2) **Right/Bottom Outer Wall:** 70.5 in, nested against the inside face of the 72.0 in outer wall so both outer sides measure 72.0 in (70.5 + 1.5 = 72.0 in). (3) **Inner Courtyard Wall B:** 49.5 in. (4) **Inner Courtyard Wall A:** 48.0 in, butted against the soil-side face of the 49.5 in board. (5 and 6) **Two Arm-End Boards:** 21.0 in each, nested between the outer and inner walls.

Shop Assembly and Squaring Sequence (See Blueprint Plate 04)

1. **Assemble the outer 72.0 by 72.0 in L-corner:** Using the staggered screw heights (1, 3, 5, 7 in vs. 2, 4, 6, 8 in), attach a 9.25 in post at the back outer corner joining the 72.0 in outer wall and the 70.5 in outer wall. Fasten a post inset 1.5 in from the far tip of both outer walls.
2. **Install arm ends and inner walls:** Fasten a post inset 1.5 in at the outer ends of the 49.5 in and 48.0 in inner courtyard walls (five posts total at the five perimeter corners; no post is used at the inner courtyard joint). Screw the two 21.0 in arm-end boards between the outer and inner walls (40 post screws total).
3. **Brace the courtyard corner (BP-04 Inset Detail):** Butt the end of the 48.0 in inner board tight against the inside face of the 49.5 in inner board and drive **three 3-in screws** through the 49.5 in board into the end of the 48.0 in board. On the exterior courtyard side, fasten the **12-gauge steel 90 degree corner brace** using six **1-1/4 in structural screws** (three per leg so screw tips never pierce the 1.5 in board).
4. **Verify squaring diagonals and reach:** Check that each 24 by 72 in arm diagonal reads **75.9 in (75-7/8 in)**, the long squaring diagonal between the two far outer corners reads **101.8 in (101-13/16 in)**, and the outer back corner to inner courtyard corner reads **33.9 in (33-7/8 in)**. From the inner courtyard corner, the far back interior post corner sits at 26.9 in (31.8 in to the interior board corner), so plant low-maintenance perennial herbs in the back corner square. Flip and staple the overlapped hardware cloth strips.

Master Bed Architecture and Volume Comparison

Bed Model	Outside Footprint	Interior Area (Gross / Net)	Soil Volume to Rim (Gross / Net)	Mid-Span Tie
4x4 Classic Bed	16.0 sq ft	14.1 sq ft / 13.7 sq ft	10.8 cu ft (0.40 cu yd) / 10.6 net	No
4x8 Workhorse Bed	32.0 sq ft	29.1 sq ft / 28.7 sq ft	22.4 cu ft (0.83 cu yd) / 22.1 net	Yes (45.0 in)
2x8 Walkway Runner	16.0 sq ft	13.6 sq ft / 13.2 sq ft	10.5 cu ft (0.39 cu yd) / 10.2 net	Yes (21.0 in)
6x6 Corner L-Bed	20.0 sq ft	17.1 sq ft / 16.6 sq ft	13.2 cu ft (0.49 cu yd) / 12.8 net	No


SPACE-SPECIALIZED GEOMETRY: RUNNER AND CORNER L-BED (BP-04)
SCALE 1:16 DUAL PLAN AND DETAIL
2x8 WALKWAY RUNNER
(24 × 96 in)
21.0 in
Mid-Span
Tie
24 in
Reach
Diag: 99.0 in | 24 in Reach
6x6 CORNER L-BED
72.0 in
21.0 in
48.0 in
49.5 in
70.5 in
21.0 in
20 sq ft Footprint | 17.1 sq ft Planting Area
Squaring Diag: 101.8 in | Arm Diag: 75.9 in
49.5 in board
48.0 in
board
12-ga Steel
Brace @
Courtyard
COURTYARD CORNER DETAIL
1:16 SCALE

Section 5: Soil Science and the Fluff Formula

COMPANION: BLUEPRINT PLATE 05 (BP-05)

Why Engineered Soil Outperforms Native Topsoil

Never fill a raised bed with native yard clay, silt, or bagged topsoil. Inside a framed enclosure, heavy topsoil packs into an airless paste when wet and bakes into a hard crust when dry. Intensive square-foot roots require an engineered growing medium with high porosity, steady moisture retention, and active soil biology, as shown in the **Blueprint Plate 05 (BP-05)** layer-by-layer cutaway.

The Fluff Formula: Equal Thirds by Volume

- **One-third coarse horticultural vermiculite:** An expanded mica mineral that holds water and dissolved nutrients inside tiny pores while keeping the mix open and aerated. Wet vermiculite lightly before pouring and wear a dust mask while mixing.
- **One-third coconut coir or sphagnum peat moss:** Provides sponge-like fibrous structure and even moisture wicking. Coir sits near an ideal pH of 5.5 to 6.8 and rewets easily. Pre-moisten sphagnum peat moss thoroughly before blending.
- **One-third multi-source blended compost:** Supplies beneficial microbes and slow-release nutrition. Blend at least three distinct organic sources, such as aged cow or horse manure, mushroom compost, worm castings, and screened leaf compost.

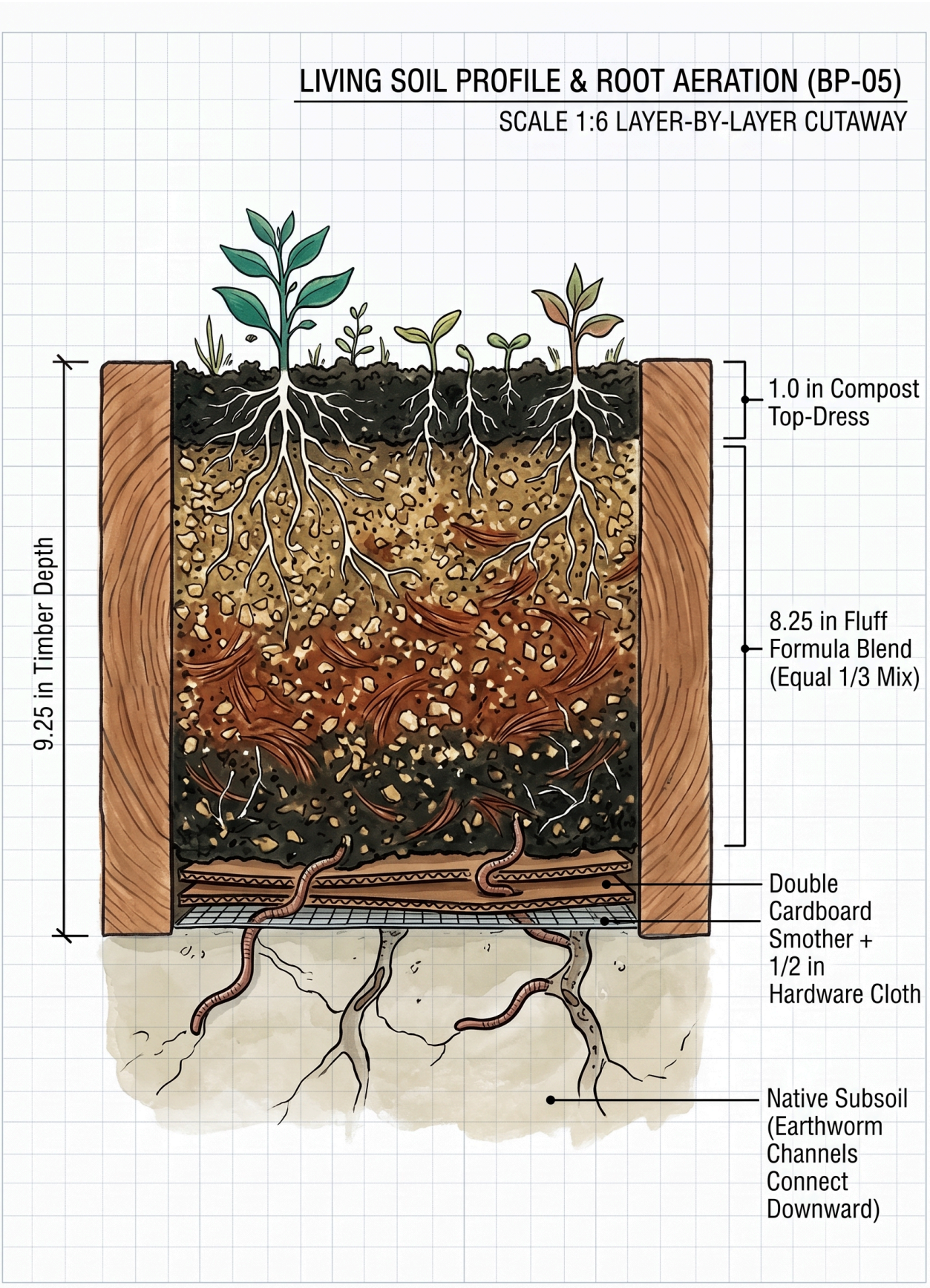
Fill Plan: 8.25 in Fluff Formula Blend Plus 1.0 in Compost Top-Dress

Following the vertical profile in **Blueprint Plate 05 (BP-05)**, fill each bed with the mixed Fluff Formula blend to within 1.0 inch of the top rim (**8.25 in depth**), then finish with a **1.0 inch layer of multi-source compost** (total 9.25 in timber depth) over a double layer of cardboard and 1/2-inch hardware cloth. In the table below, exact figures appear alongside rounded purchasing numbers in parentheses (3 x Each Third = Blend, and Blend + Top-Dress = Total to Rim). Because dry components settle 10 to 15 percent when watered and cardboard/posts displace about 0.25 to 0.33 cu ft, purchase roughly 10 percent extra bulk volume.

Bed Model	Gross Interior	Blend (8.25 in)	Each Third (1/3)	Top-Dress (1 in)	Total to Rim (Net)
4x4 Classic	45 x 45 in (14.06 sq ft)	9.67 cu ft (9.7)	3.22 cu ft (3.2)	1.17 cu ft (1.2)	10.84 cu ft (10.59 net)
4x8 Workhorse	45 x 93 in (29.06 sq ft)	19.98 cu ft (20.0)	6.66 cu ft (6.7)	2.42 cu ft (2.4)	22.40 cu ft (22.07 net)
2x8 Runner	21 x 93 in (13.56 sq ft)	9.32 cu ft (9.3)	3.11 cu ft (3.1)	1.13 cu ft (1.1)	10.45 cu ft (10.17 net)
6x6 L-Bed	2,457 sq in (17.06 sq ft)	11.73 cu ft (11.7)	3.91 cu ft (3.9)	1.42 cu ft (1.4)	13.15 cu ft (12.82 net)

CARDBOARD GROUND SMOTHER AND NO-TILL SPRING TOP-OFF PROTOCOL

Mow turf short, set the mesh-lined frame on top, and lay two overlapping layers of plain corrugated cardboard (all plastic tape, staples, and glossy ink removed) across the floor with 6-inch seam overlaps. Soak thoroughly with a hose before adding soil so earthworms can digest the cardboard and open channels downward into the native subsoil. Once filled, never dig or rototill the bed: simply top off with 1.0 inch of fresh multi-source compost every spring to restore nutrients and biology naturally.


LIVING SOIL PROFILE & ROOT AERATION (BP-05)
SCALE 1:6 LAYER-BY-LAYER CUTAWAY
9.25 in Timber Depth
1.0 in Compost Top-Dress
8.25 in Fluff Formula Blend (Equal 1/3 Mix)
Double Cardboard Smother + 1/2 in Hardware Cloth
Native Subsoil (Earthworm Channels Connect Downward)

Section 6: Layout Blueprint 01, 16-Cell Kitchen Salad Box

COMPANION: BLUEPRINT PLATE 06 (BP-06)

Mapped visually in **Blueprint Plate 06 (BP-06)**, this plan fits a **4x4 Classic Bed** (sixteen 11.25 by 11.25 in cells) outside the kitchen door. Rows run east to west, with Row A along the north trellis axis and Row D along the sunny southern edge, so tall crops never shade low salad greens. Each cell follows a three-window succession schedule assuming a last spring frost in late April and a first fall frost in late October (shift dates to match your local frost calendar). Because no alliums appear in this bed, the Section 10 legume-and-allium separation rule is automatically satisfied.

Three-Season 16-Cell Succession Schedule (Rows A to D)

Cells	Spring (Mar to May)	Summer (Jun to Aug)	Fall (Sep to Nov)
A1, A4	Sugar snap peas, 8/sq, on twine (sow Mar 15)	Pole beans, 4/sq, on twine (sow Jun 10 after peas)	Pole beans until first frost
A2, A3	Radishes, 16/sq (sow Apr 1, pull by May 10)	Indeterminate cherry tomatoes, 1/sq single stem on stakes (transplant May 15)	Cherry tomatoes until first frost
B1	Lacinato kale, 1/sq (transplant Mar 25)	Lacinato kale (continuous outer-leaf harvest)	Kale (leaves sweeten after light frost)
B2, B3	Butterhead lettuce, 4/sq (sow Mar 15)	Genovese basil, 4/sq (transplant May 20)	Basil harvest until first frost
B4	Butterhead lettuce, 4/sq (sow Mar 15)	Swiss chard, 4/sq (sow May 20)	Swiss chard until hard frost
C1	Flat-leaf parsley, 4/sq (transplant Apr 1)	Flat-leaf parsley (steady stem harvest)	Parsley into late autumn
C2, C3	Baby spinach, 9/sq (3x3 grid, sow Mar 15)	Bush beans, 4/sq (sow May 20, harvest Jul 10 to Aug 15)	Baby spinach, 9/sq (sow Sep 1)
C4	Red leaf romaine, 4/sq (sow Mar 15)	Bush beans, 4/sq (sow May 20, clear by Aug 20)	Red leaf romaine, 4/sq (sow Aug 25)
D1	Nantes carrots, 16/sq (sow Apr 1, harvest Jun 15)	Rest under straw mulch (Jun 15 to Jul 31)	Nantes carrots, 16/sq (sow Aug 1, harvest Oct 20)
D2, D3	French breakfast radishes, 16/sq (sow Apr 1)	Table beets, 9/sq (sow Jun 1, pull by Aug 25)	French breakfast radishes, 16/sq (sow Sep 1)
D4	Radishes, 16/sq (sow Apr 15, pull by May 15)	French marigold, 1/sq (transplant May 15)	Marigold blooms until frost

SINGLE-STEM TOMATO PRUNING AND CUT-AND-COME-AGAIN HARVESTING

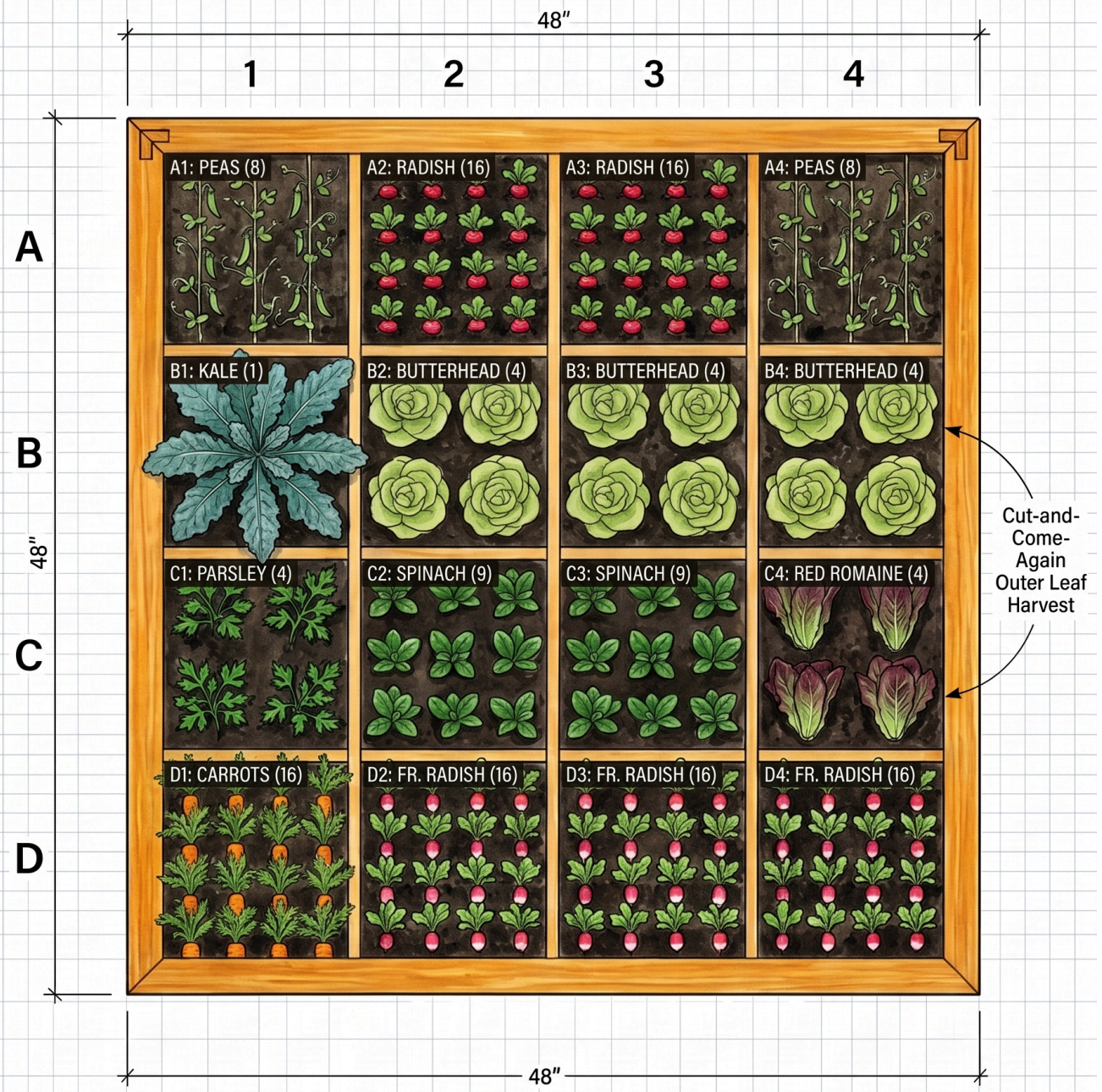
1. Tomato suckers (Cells A2 and A3): Train indeterminate cherry tomatoes to one central stem. Every week, pinch off axillary side shoots (suckers) in the leaf crotches while under 2 inches long. Tie the central leader to its stake every 8 to 12 inches of height so airflow stays open and plant energy goes into early fruit clusters.

2. Cut-and-come-again greens: Do not pull whole plants of loose-leaf lettuce, kale, chard, or spinach during their growth window. Snip mature outer leaves about 1 inch above the soil line and leave the central crown intact so plants keep producing fresh leaves for weeks.

16-CELL KITCHEN GARDEN PLANTING MAP (BP-06)

SCALE 1:12 • SPRING PLANTING WINDOW & ROTATIONS

▲ NORTH TRELLIS AXIS (ROW A)



SUMMER SWAPS: A2-A3 → Cherry Tomatoes (1) | B2-B3 → Basil (4) | B4 → Chard (4) | C2-C4 → Bush Beans (4) | D2-D3 → Beets (9) | D4 → Marigold (1)

Section 7: Layout Blueprint 02, 32-Cell Summer Feast

COMPANION: BLUEPRINT PLATE 07 (BP-07)

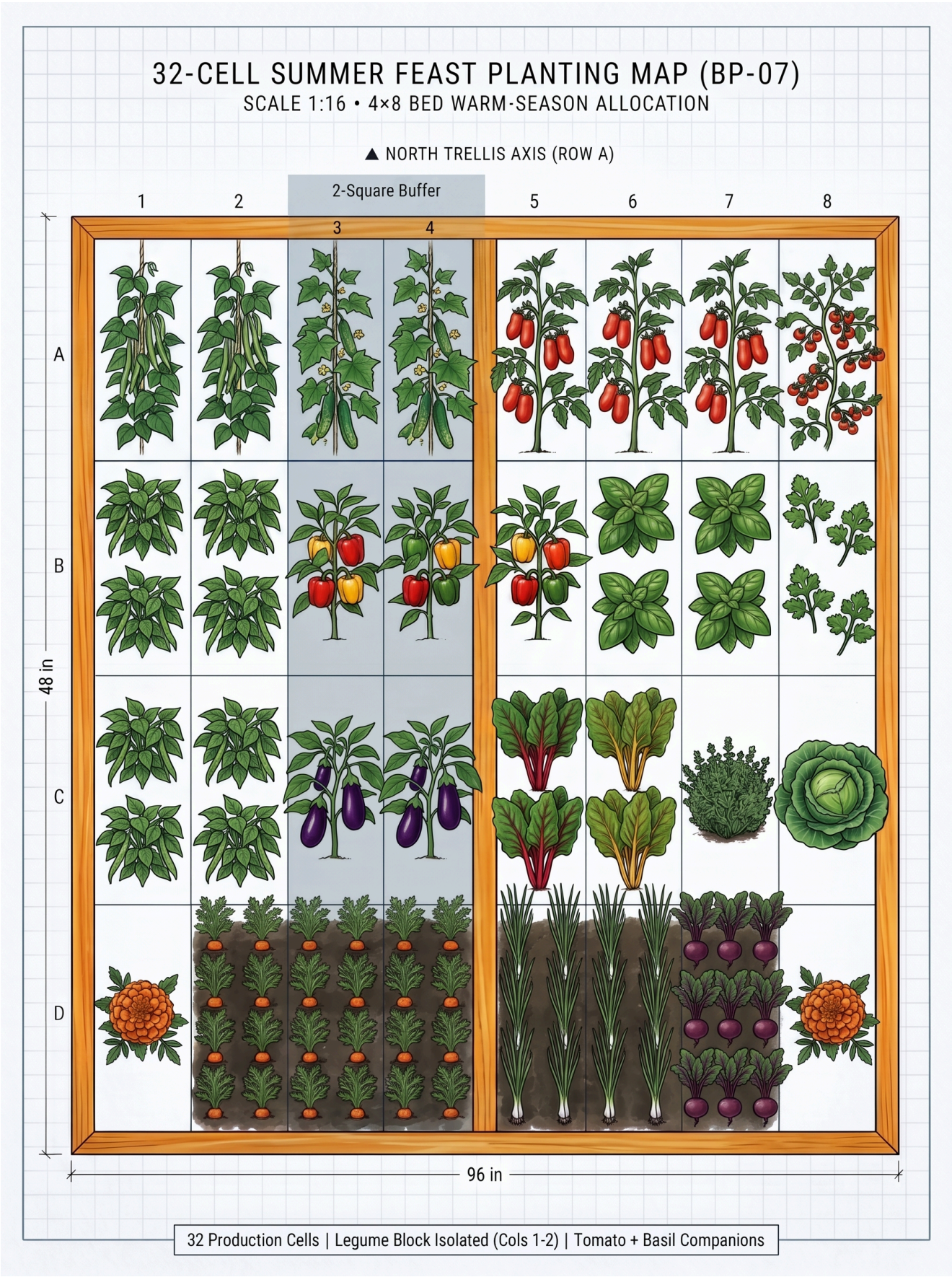
Illustrated in **Blueprint Plate 07 (BP-07)**, this panoramic layout covers the **4x8 Workhorse Bed** (thirty-two cells of 11.25 by 11.625 in) for the warm-season window from mid-May through autumn frost. Columns run **1 to 8 from west to east** and Rows **A to D from north to south**. All beans and peas are grouped on the west end in Columns 1 and 2, while scallions sit in Columns 5 and 6 (D5, D6), maintaining a strict **two-column (2-Square Buffer)** across Columns 3 and 4 between legumes and alliums.

Panoramic 32-Cell Warm-Season Allocation (Columns 1 to 8, Rows A to D)

Cells	Crop and Cultivar	Per Sq	Placement and Succession Notes
A1, A2	Kentucky Wonder pole beans	4/sq	Sown Jun 10 on twine drop after spring snap peas (sown Mar 15) finish
A3, A4	Marketmore 76 slicing cucumbers	2/sq	Trained up northern trellis; part of Columns 3 to 4 buffer zone
A5, A6, A7	San Marzano paste tomatoes	1/sq	Single stem on overhead twine (transplant May 10)
A8	Indeterminate cherry tomato	1/sq	Single stem on twine at northeast corner (transplant May 10)
B1, B2	Bush snap beans (Provider)	4/sq	Grouped with C1, C2 and A1, A2 in west-end legume block
B3, B4, B5	Sweet bell peppers (California Wonder)	1/sq	Middle solar tier; B3 and B4 buffer legumes; B5 sits south of tomato A5
B6, B7	Genovese basil	4/sq	Full sun directly south of paste tomatoes A6 and A7
B8	Curly parsley	4/sq	Directly south of cherry tomato A8 along eastern edge
C1, C2	Bush green beans (Provider)	4/sq	Grouped with B1, B2 legume block; harvest Jul 10 to Aug 15
C3, C4	Japanese eggplants	1/sq	Same nightshade family as tomatoes/peppers; rotate per Section 11
C5, C6	Swiss chard (Bright Lights)	4/sq	Continuous outer-leaf harvest through autumn
C7	Culinary thyme	1/sq	Low woody perennial herb
C8	Golden Acre cabbage, then baby spinach	1, then 9/sq	Spring cabbage (Apr 1 to Jul 10), followed by fall spinach (sown Sep 1)
D1, D8	French marigold (Tagetes patula)	1/sq	Southwest (D1) and Southeast (D8) front corner pollinator flowers
D2, D3, D4	Danvers half-long carrots	16/sq	Sown Apr 1 and Aug 1; D4 adjoins scallions in D5
D5, D6 / D7	D5 to D6: Scallions | D7: Table beets	16/sq | 9/sq	D5 to D6: Scallions (16/sq, 2 cols from beans). D7: Table beets (9/sq, 3x3 grid)

TOMATO PRUNING AND CENTER CROSS-TIE CLEARANCE (COLUMNS 4 AND 5)

Prune indeterminate tomatoes (A5 to A8) to a single stem, pinching axillary suckers weekly and tying leaders to twine drops every 8 to 12 inches. The recessed 2x4 cross-tie runs between Columns 4 and 5 at 4.0 inches below the top rim, holding the walls parallel against lateral soil load while leaving upper planting depth completely clear across the center.


32-CELL SUMMER FEAST PLANTING MAP (BP-07)
SCALE 1:16 • 4×8 BED WARM-SEASON ALLOCATION
▲ NORTH TRELLIS AXIS (ROW A)
2-Square Buffer
1
2
3
4
5
6
7
8
A
B
C
D
48 in
96 in
32 Production Cells | Legume Block Isolated (Cols 1-2) | Tomato + Basil Companions

Section 8A: Layout Blueprint 03, 2x8 Vertical Fence Bed

COMPANION: BLUEPRINT PLATE 08 (BP-08)

A narrow fence-line bed captures high solar potential from two tiers: climbing vines trained up the back fence and compact greens and root crops along the open front pathway. Build the bed as the **2x8 Walkway Runner** in Section 4A (interior 21.0 by 93.0 in, sixteen cells of 10.5 by 11.625 in, 13.6 sq ft gross interior area). As shown in **Blueprint Plate 08 (BP-08)**, orient the bed along a south-facing fence (meaning the fence stands along the northern edge of the bed and its open face looks south), with the rear timber wall set 2 to 3 inches from the fence face.

Two-Tier 16-Cell Fence Runner Allocation (Columns 1 to 8)

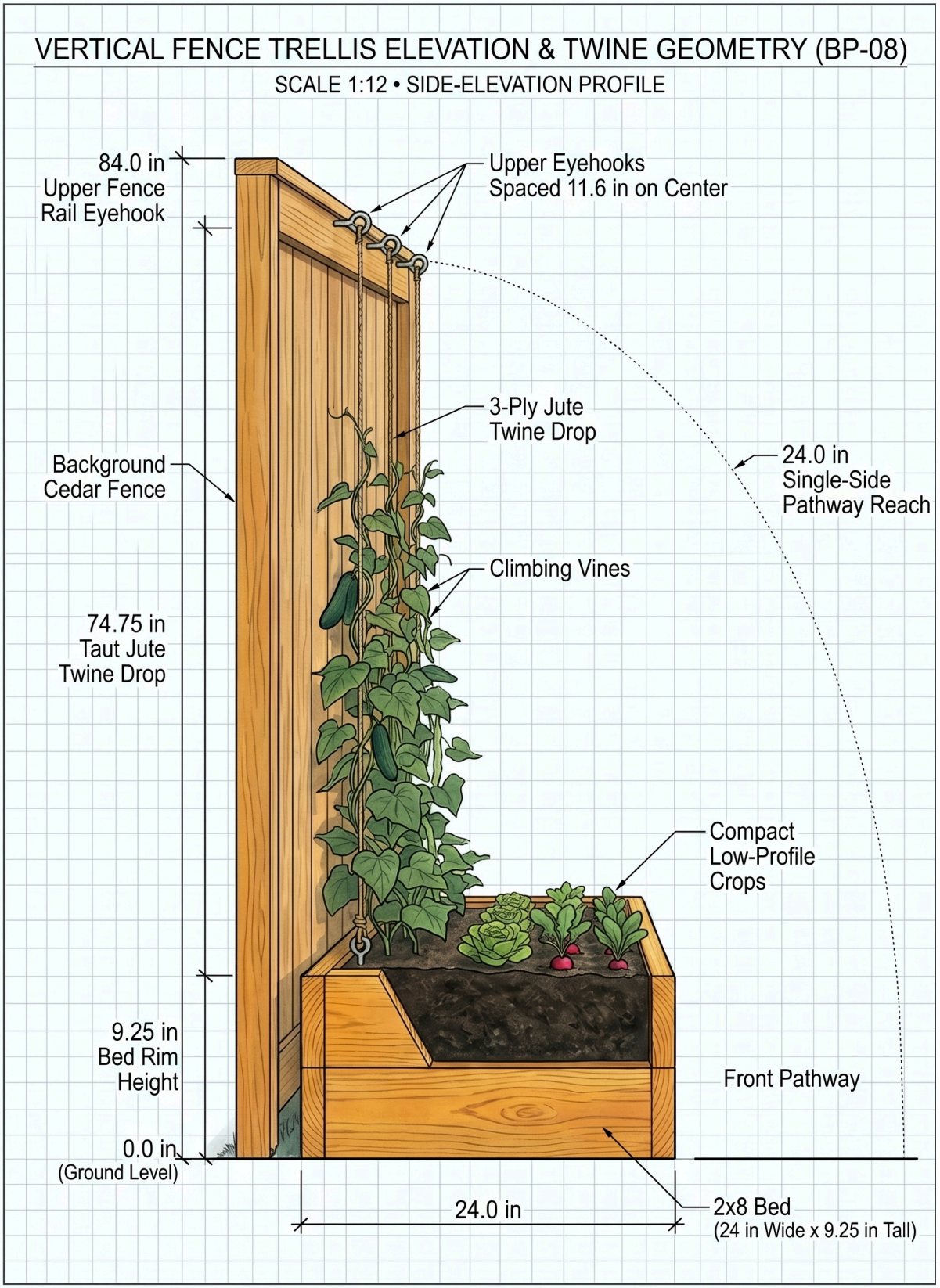
Tier	Cells	Crop and Seasonal Succession	Per Sq
Back (north fence)	1 to 4	Marketmore 76 slicing cucumbers (vertical jute twine trellis)	2 per sq
Back (north fence)	5 to 8	Kentucky Wonder climbing pole beans, trained to upper fence rail	4 per sq
Front (south edge)	1	Flowering dill, beside the cucumbers to attract aphid-eating hoverflies	4 per sq
Front (south edge)	2 to 4	Loose-leaf lettuce (spring/fall), followed by Swiss chard (summer)	4 per sq
Front (south edge)	5 to 6	French breakfast radishes (spring/fall), followed by Table beets at 9/sq (summer)	16 (9)
Front (south edge)	7 to 8	Baby spinach (spring/fall), followed by Table beets at 9/sq (summer)	9 per sq

Overhead Eyehook and Jute Twine Dropline System (See Blueprint Plate 08)

1. **Install upper rail hooks:** Referring to the side-elevation profile in **Blueprint Plate 08 (BP-08)**, drive eight 3-in zinc screw eyehooks into a structural horizontal upper fence rail (never into thin fence pickets) at **84.0 in height** on a 7-ft fence. Space hooks **11.625 in apart on center** (11.6 in rounded on diagram), starting 5.81 in from the inside end wall so each hook sits directly above the center of a back-tier cell (cells 1 to 8).
2. **Install lower rim hooks:** Pre-drill and screw a matching eyehook vertically into the top edge of the rear 2x10 timber wall (**9.25 in Bed Rim Height** above ground level), directly plumb below each upper hook.
3. **String taut drops:** Tie lengths of **3-ply jute twine** across the **74.75 in vertical span** (84.0 minus 9.25 equals 74.75 in; cut about 90 in per line to allow for knots) and pull taut.
4. **Train vines upward:** Cucumbers climb via tendrils; tie heavy stems loosely every 8 to 12 inches. Pole beans have no tendrils: their shoots twine counterclockwise as viewed from above (wrapping from left to right across the front of the twine as they climb), so guide young runners in that direction.
5. **Autumn cleanup:** Cut the biodegradable jute twine at both ends after frost and compost cords and healthy spent vines together.

CRAFTSMAN HARDWARE TIP: MASONRY OR METAL FENCES

If your northern boundary is brick, concrete block, or metal panel where screw eyehooks cannot bite, drive two 8-foot steel T-posts behind the bed ends and bolt a single 8-foot 2x4 horizontal top rail across them at 84.0 inches height.


VERTICAL FENCE TRELLIS ELEVATION & TWINE GEOMETRY (BP-08)
SCALE 1:12 • SIDE-ELEVATION PROFILE
84.0 in
Upper Fence
Rail Eyehook
Upper Eyehooks
Spaced 11.6 in on Center
3-Ply Jute
Twine Drop
24.0 in
Single-Side
Pathway Reach
Background
Cedar Fence
Climbing Vines
74.75 in
Taut Jute
Twine Drop
Compact
Low-Profile
Crops
9.25 in
Bed Rim
Height
Front Pathway
0.0 in
(Ground Level)
24.0 in
2x8 Bed
(24 in Wide x 9.25 in Tall)

Section 8B: Architectural Extension, 6x6 Corner L-Bed Courtyard Matrix

COMPANION: BLUEPRINT PLATE 04 (BP-04)

> **FRAME REFERENCE: BLUEPRINT PLATE 04 (PAGE 8)**
> **Frame Reference:** This layout maps directly onto the 6x6 Corner L-Bed timber frame detailed in Blueprint Plate 04 (Page 8). Refer to Blueprint Plate 04 for exact board joinery, dimensions, and courtyard steel brace assembly.

To ensure every timber frame in this bundle includes a complete planting plan, this extension maps all **19 workable squares** of the **6x6 Corner L-Bed** (17.1 sq ft planting area). Site the L-Bed in a sunny northern corner so the 72.0 in North Arm runs east to west along the back and the 72.0 in Side Arm extends southward around the 48.0 by 48.0 in walk-in courtyard.

Courtyard Corner L-Bed Allocation (North Arm, Back Corner, and Side Arm)

L-Bed Zone	Squares	Crop and Three-Season Role	Density
Deep Back Corner	Corner 1 (Outer)	**Perennial Rosemary and Winter Savory** (longest reach at 26.9 to 31.8 in; low maintenance)	1 per sq
Deep Back Corner	Corner 2 (Inner)	**Perennial English Thyme and Oregano** (step-in courtyard access; woody culinary herbs)	1 to 2 / sq
North Arm (Back Tier)	N-Back 1 to 4	Indeterminate cherry and Juliet plum tomatoes on stakes (set May 15 after spring radishes)	1 per sq
North Arm (Front Tier)	N-Front 1 to 2	Genovese basil (summer) following spring butterhead lettuce (sown Mar 15)	4 per sq
North Arm (Front Tier)	N-Front 3 to 4	Scallions (16/sq) and Danvers carrots (16/sq) along sunny inner courtyard rim	16 per sq
Side Arm (Outer Tier)	S-Outer 1 to 2	California Wonder bell peppers (1/sq, medium height), buffered 3+ squares from scallions	1 per sq
Side Arm (Outer Tier)	S-Outer 3 to 4	Provider bush green beans (4/sq, summer) between spring and fall baby spinach (9/sq)	4 / sq (9)
Side Arm (Inner Tier)	S-Inner 1 to 3	Cut-and-come-again salad bar: Red romaine, arugula, and Bright Lights Swiss chard	4 per sq
Side Arm (Front Tip)	S-Inner 4	French marigold and trailing sweet alyssum (courtyard entrance pollinator habitat)	1 to 2 / sq

Why the L-Bed Excels as a Courtyard Herb and Salad Garden

- **Smart use of the deep corner:** As noted in Section 4B, the far back interior post sits 26.9 inches from the inner courtyard corner (31.8 in to the back board corner). Planting woody perennials like rosemary, thyme, and oregano in the deep corner squares means you only reach in for occasional harvest snips rather than weekly seeding.
- **Courtyard microclimate and legume buffer:** Standing inside the 48.0 by 48.0 in courtyard puts all inner-tier greens within easy reach, while scallions in N-Front 3 sit more than 48 inches away from bush beans in S-Outer 3 to 4.

Section 9: Companion Planting Science

COMPANION: BLUEPRINT PLATE 09 (BP-09)

In bio-intensive beds, roots and foliage intermingle closely, so neighboring crops matter. Rather than repeating garden folklore, **Blueprint Plate 09 (BP-09)** illustrates four field-researched interactions supported by three practical mechanisms: volatile aromatic compounds that help mask host odors, compatible canopy light sharing, and floral nectar that recruits predatory and parasitic insects. Because field results vary with weather and local pest pressure, treat companion pairings as helpful habitat support rather than standalone pest control.

Evidence-Based Companion Planting Matrix (See Blueprint Plate 09)

Primary Crop	Companion	Proposed Mechanism	Target Pest	Evidence Rating
01 / Tomato (*Solanum lycopersicum*)	Genovese basil (*Ocimum basilicum*)	Basil emits volatile terpenoids/phenylpropanoids (linalool, eugenol) that may mask host scent	Hornworm moths, whiteflies	**[EVIDENCE: LIMITED]** Culinary and sun match, not standalone control
02 / Carrot	Scallions, leeks (alliums)	Allium sulfur volatiles may disrupt olfactory host finding by egg-laying female rust flies	Carrot rust fly (*Psila rosae*)	**[EVIDENCE: MIXED]** Helpful deterrent; insect row cover is more reliable
03 / Cucumber	Flowering dill	Shallow yellow umbel florets supply nectar and pollen to adult syrphid hoverflies and predatory wasps	Aphids (*Aphis gossypii*)	**[EVIDENCE: GOOD]** Hoverfly larvae feed heavily on aphids
04 / Brassicas (Kale, Cabbage)	Sweet alyssum (*Lobularia maritima*)	Continuous floral nectar supports hoverflies and braconid wasps (*Cotesia glomerata, C. rubecula*)	Cabbageworms, cabbage aphids	**[EVIDENCE: GOOD]** Strong evidence for natural enemy recruitment

Four Practical Field Notes on These Pairings

- **Tomatoes and basil:** Both thrive in warm soil and full sun. Place basil directly south of staked tomatoes (such as B6 to B7 below A6 to A7 in the 32-cell plan) so it receives full southern light without competing for trellis space.
- **Carrots and alliums:** Set carrots and scallions in adjoining southern squares (D2 to D4 beside D5 to D6). Where carrot rust fly pressure is severe, cover carrot squares with fine insect mesh row cover at sowing.
- **Cucumbers and flowering dill:** Allow a few dill stems in front cell 1 to bolt into yellow flower umbels. Hoverfly larvae prey heavily on soft-bodied aphids (note that hoverfly larvae do not control spider mites).
- **Brassicas and sweet alyssum:** Tuck low-growing sweet alyssum along bed corners beside kale or cabbage to keep beneficial *Cotesia* wasps foraging all season.

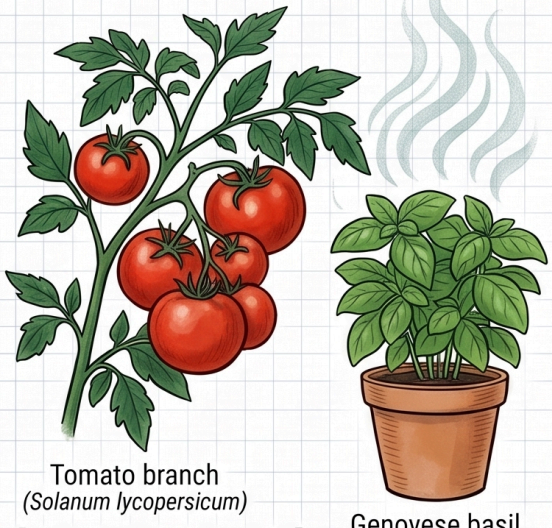
A REALISTIC SCIENTIFIC NOTE ON FRENCH MARIGOLDS (TAGETES PATULA)

Certain French marigold cultivars (*Tagetes patula*) release root alpha-terthienyl, which suppresses root-knot nematodes (*Meloidogyne* spp.) only when grown as a dense, solid cover crop for a full season before planting susceptible crops. A single marigold in a corner cell will not suppress root-knot nematodes. In this bundle, marigolds are planted strictly for pollinator attraction and beneficial insect bloom habitat.

COMPANION PLANTING MATRIX (BP-09)

EVIDENCE-BASED BENEFICIAL INSECT & AROMATIC INTERACTIONS

01 / TOMATOES + BASIL

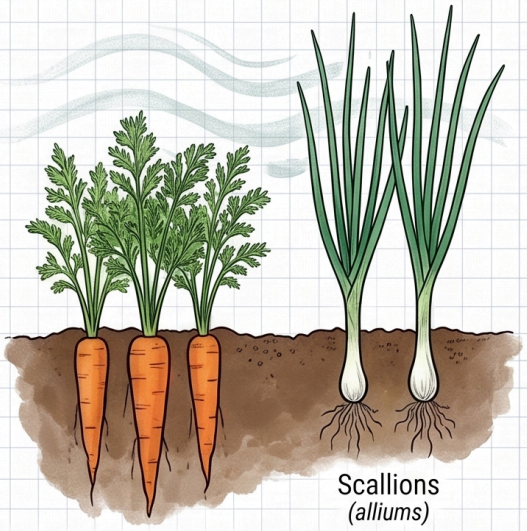


Tomato branch
(Solanum lycopersicum)

Genovese basil
(Ocimum basilicum)

[EVIDENCE: LIMITED]
Culinary & sun match

02 / CARROTS + ALLIUMS



Scallions
(alliums)

[EVIDENCE: MIXED]
Rust fly deterrence (Row cover recommended)

03 / CUCUMBERS + FLOWERING DILL

[EVIDENCE: GOOD]
Hoverfly larvae feed on aphids

04 / BRASSICAS + SWEET ALYSSUM

[EVIDENCE: GOOD]
Nectar recruits Cotesia wasps

0 1:1

1:1 SCALE MATRIX (NTS)

SHEET BP-09

Section 10: Crop Defense and Pest Traps

COMPANION: BLUEPRINT PLATE 10 (BP-10)

Four Biological Crop Clashes to Avoid (See Blueprint Plate 10)

Where roots and foliage touch inside an intensive grid, certain neighbors share destructive pathogens or inhibit growth. Avoid the four combinations diagrammed in the upper panels of **Blueprint Plate 10 (BP-10)**:

- **1. Tomatoes vs. Potatoes (Shared Late Blight):** Both belong to the nightshade family (*Solanaceae*) and share early blight (*Alternaria solani*), late blight (*Phytophthora infestans*), and Colorado potato beetles. Keep them in separate beds or grow potatoes in dedicated containers, and promptly pull any volunteer potato shoots that sprout from compost.
- **2. Florence or Herb Fennel in the Bed (Allelopathic Stunting):** Fennel (*Foeniculum vulgare*) is a vigorous root competitor that releases allelopathic compounds capable of stunting bush beans, tomatoes, and brassicas. Do not plant fennel in raised vegetable beds; grow it in a standalone container only.
- **3. Black Walnut Juglone (Root Toxicity Zone):** Black walnut (*Juglans nigra*) roots, hulls, and leaves exude juglone, a naphthoquinone compound toxic to nightshades (tomatoes, peppers, eggplants), potatoes, and many greens. Site raised beds at least **50 to 80 feet away** from the tree's drip line, line the bed floor over an impermeable barrier if closer, and never use walnut leaves or wood chips as mulch.
- **4. Legumes vs. Alliums (Precautionary 2-Square Buffer):** Traditional garden observation holds that onions, scallions, and garlic can inhibit peas and beans. Controlled scientific evidence is limited, but keeping a **two-square (24-inch) buffer** (two 12-inch buffer cells between bean cells and scallion cells, as shown in **Blueprint Plate 10** and Section 7) is an easy, zero-cost precaution.

The Sacrificial Nasturtium Aphid Trap System (See Blueprint Plate 10)

Black bean aphids and green peach aphids frequently colonize tender shoots on beans, tomatoes, and kale. Instead of applying broad-spectrum insecticides that wipe out ladybeetles and hoverflies, use trailing nasturtiums (*Tropaeolum majus*) as an external perimeter trap crop as shown at the bottom of **Blueprint Plate 10 (BP-10)**:

1. **Set a 12-inch container 4 feet outside the bed perimeter:** Plant three trailing nasturtiums in a standalone 12-inch terracotta or resin pot placed about **4 feet (~1.2 m) outside the raised timber bed perimeter**. Do not plant trap nasturtiums inside the bed itself, where crawling wingless aphids can walk directly onto vegetable stems.
2. **Scout weekly and rinse or bag damaged stems:** Check the underside of nasturtium leaves once a week. When dark aphid clusters build up on a stem, snip the infested stems directly into a sealed trash bag or rinse them off away from the garden, leaving beneficial predator eggs on lightly colonized leaves.
3. **Scout main beds concurrently:** A trap pot reduces (but does not eliminate) incoming winged aphids. Check main crop tips weekly and knock off early colonizers with a firm stream of water or potassium-salt insecticidal soap.

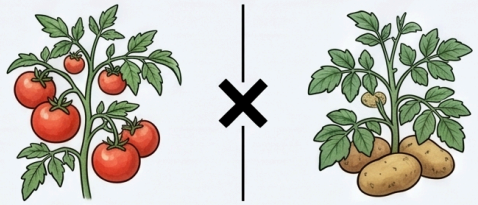
> **FIELD HYGIENE REMINDER: KEEP BLIGHTED AND INFESTED FOLIAGE OUT OF HOME COMPOST**
> Never toss heavily aphid-infested nasturtium trimmings or blighted nightshade foliage into a backyard compost pile, which rarely heats uniformly enough (140°F+) to kill spores and pests. Seal them into municipal waste bags to prevent re-infesting adjacent beds.

CROP CLASHES & PEST DEFENSE BLUEPRINT (BP-10)

SCALE 1:1 • BIOLOGICAL INCOMPATIBILITIES & TRAP CROP DYNAMICS

TOMATOES vs POTATOES

Shared Late Blight (Keep in Separate Beds)

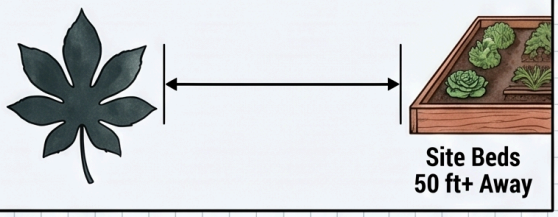


FENNEL IN THE BED

Allelopathic Growth Stunting (Container Only)

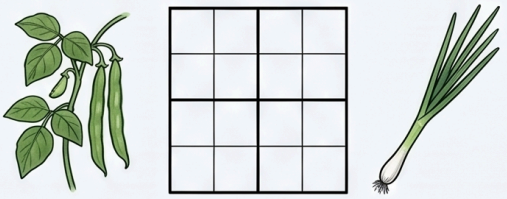
BLACK WALNUT JUGLONE

Root Toxicity Zone (Site Beds 50 ft+ Away)

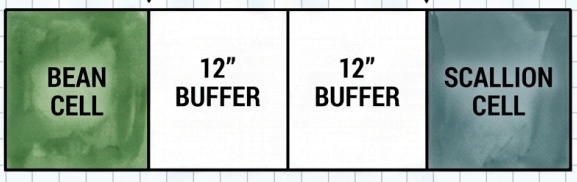


LEGUMES vs ALLIUMS

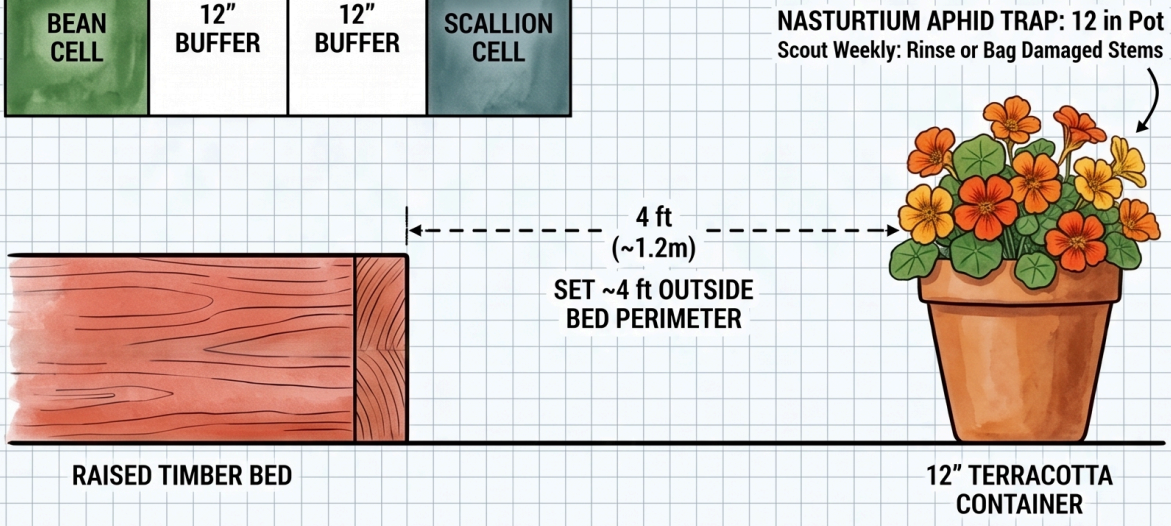
Growth Inhibition (Precautionary 2-Square Buffer)



PRECAUTIONARY 2-SQUARE BUFFER



SACRIFICIAL NASTURTIUM APHID TRAP SYSTEM & SPACING



1:1 WARNING & TRAP SCHEMATIC | SHEET BP-10

Section 11: Seasonal Succession and Rotations

COMPANION: BLUEPRINT PLATE 11 (BP-11)

Three-Season Temperature Windows (See Blueprint Plate 11 Outer Ring)

A raised bed stays productive for nine to ten months when succession plantings follow the three seasonal temperature windows illustrated on the outer ring of **Blueprint Plate 11 (BP-11)**. Air temperatures and dates below assume a last spring frost in late April and a first fall frost in late October; adjust by the number of weeks your local frost dates differ:

- **Window 1: Cool Spring (45 to 65°F, March to mid-May):** Sow or transplant radishes, baby spinach, sugar snap peas, lettuce, kale, cabbage, carrots, scallions, and chard. Fast crops (radishes before May 15 tomatoes; spinach before May 20 bush beans) clear their squares by mid-May; snap peas finish by Jun 10 for pole beans, and spring cabbage clears by Jul 10.
- **Window 2: Warm Season (70 to 90°F, mid-May to early September):** After the last frost, with soil at least **60°F** (65°F for peppers and eggplants), transplant tomatoes, peppers, eggplants, and basil, and direct-sow bush beans, pole beans, and cucumbers under a full sun canopy. Indeterminate tomatoes and peppers produce until frost.
- **Window 3: Crisp Autumn (50 to 68°F, August to November):** As short-season summer crops finish (bush beans in mid-August, summer beets in late August), sow fall carrots (Aug 1), red romaine (Aug 25), and fall spinach and radishes (Sep 1). Light autumn frosts convert starches to sweet sugars in kale, carrots, and spinach.

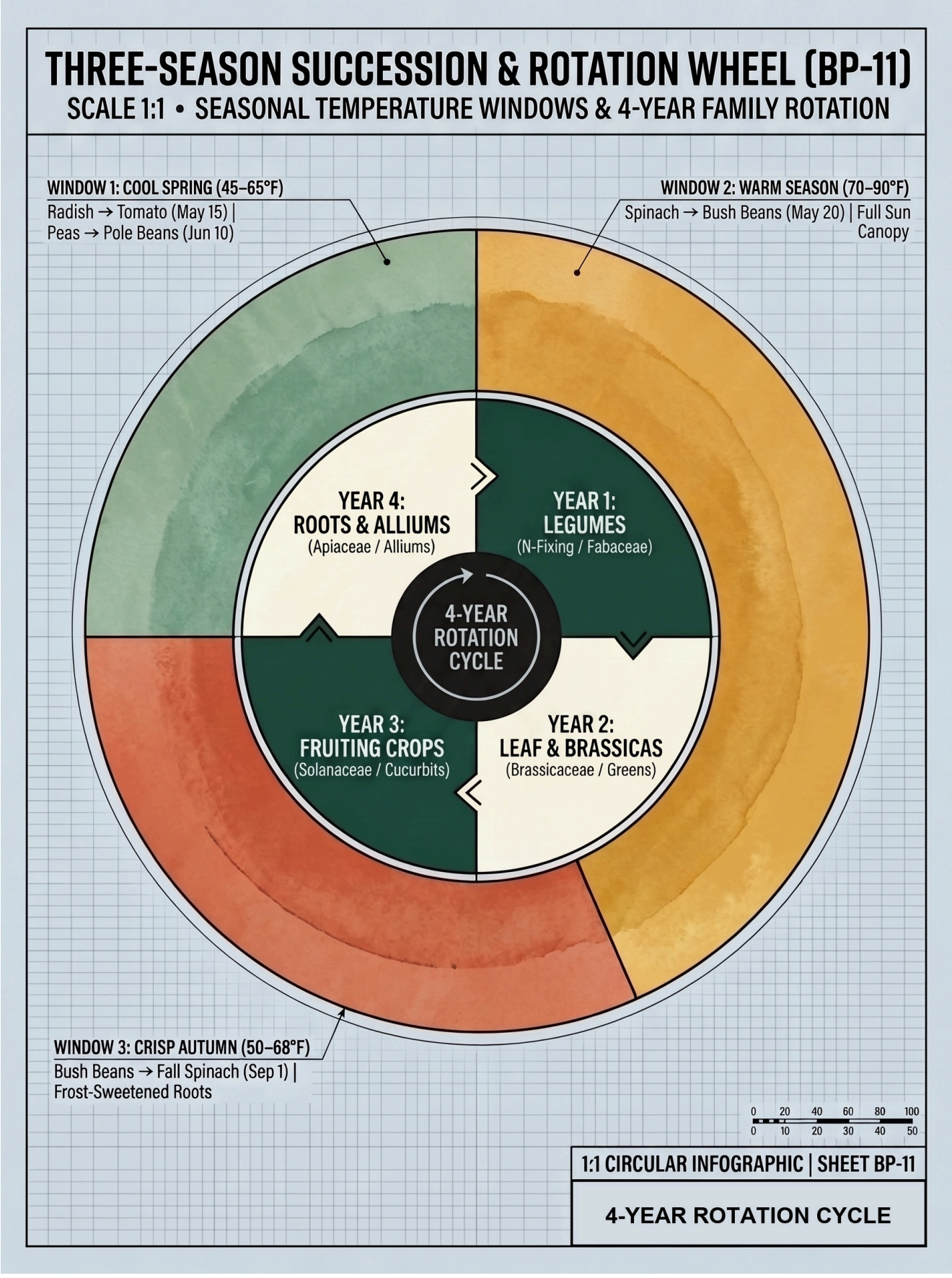
Four-Year Rotation by Plant Family (See Blueprint Plate 11 Inner Wheel)

Following the 4-Year Rotation Cycle in the center of **Blueprint Plate 11 (BP-11)**, rotating botanical families across cells prevents soil-borne pathogens from building up and balances nutrient draw. Note that radishes, cabbage, and kale all belong to *Brassicaceae* and rotate together as one group:

Year	Rotation Group	Botanical Families	Representative Crops	Soil Role
Year 1	**Legumes**	*Fabaceae*	Snap peas, bush beans, pole beans	Host nitrogen-fixing rhizobia in root nodules
Year 2	**Leaf and Brassicas**	*Brassicaceae, Amaranthaceae, Asteraceae*	Cabbage, kale, radishes, spinach, chard, beets, lettuce	Leafy nitrogen consumers
Year 3	**Fruiting Crops**	*Solanaceae, Cucurbitaceae*	Tomatoes, peppers, eggplants, cucumbers	Heavy P/K and calcium feeders; top-dress compost
Year 4	**Roots and Alliums**	*Apiaceae, Amaryllidaceae*	Carrots, parsnips, parsley, dill, scallions, leeks, chives	Lighter feeders; break up lower soil profile

ROTATING WITHIN SOLAR TIERS WITHOUT LOSING SUN ACCESS

Rotate cell blocks within each height zone so tall crops remain along the north edge. In Row A of your 4x8 bed, shift the pole bean/pea block (A1 to A2), cucumber block (A3 to A4), and tomato block (A5 to A8) horizontally each year, and rotate nightshade blocks (tomatoes, peppers, eggplants) out of the same cells for at least three years to prevent soil-borne disease buildup.


THREE-SEASON SUCCESSION & ROTATION WHEEL (BP-11)
SCALE 1:1 • SEASONAL TEMPERATURE WINDOWS & 4-YEAR FAMILY ROTATION
WINDOW 1: COOL SPRING (45–65°F)
Radish → Tomato (May 15) | Peas → Pole Beans (Jun 10)
WINDOW 2: WARM SEASON (70–90°F)
Spinach → Bush Beans (May 20) | Full Sun Canopy
YEAR 4: ROOTS & ALLIUMS
(Apiaceae / Alliums)
YEAR 1: LEGUMES
(N-Fixing / Fabaceae)
4-YEAR ROTATION CYCLE
YEAR 3: FRUITING CROPS
(Solanaceae / Cucurbits)
YEAR 2: LEAF & BRASSICAS
(Brassicaceae / Greens)
WINDOW 3: CRISP AUTUMN (50–68°F)
Bush Beans → Fall Spinach (Sep 1) | Frost-Sweetened Roots
0 20 40 60 80 100
0 10 20 30 40 50
1:1 CIRCULAR INFOGRAPHIC | SHEET BP-11
4-YEAR ROTATION CYCLE

Section 12: Field Workbook and Maintenance Protocols

COMPANION: BLUEPRINT PLATE 12 (BP-12)

MORNING SOIL-LEVEL IRRIGATION AND LATE-JUNE CLEAN STRAW MULCHING

1. Water 6:00 to 9:00 AM at soil level: Use a gentle wand or drip line at the base of stems when the top 1 to 2 inches feel dry; keeping foliage dry prevents early blight and bacterial spot.

2. Late-June mulch and herbicide check: Once soil is warm in late June, spread 2 to 3 inches of clean, seed-free wheat straw or shredded leaves across bare soil (leaving a 1-inch ring clear around stems). **Herbicide warning:** Verify that straw, hay, or manure never came from pastures treated with persistent pyridine herbicides (*aminopyralid, clopyralid, or picloram*), which can severely curl and ruin tomatoes, beans, and peas.

Complete Master Field Planting Record Worksheet (All 32 Cells, Ref: BP-12)

Every cell from the Section 7 32-Cell Summer Feast plan (A1 to D8) is accounted for below to pair with your printable **Blueprint Plate 12 (BP-12)** Master 32-Cell Field Planning Worksheet:

Bed Cell ID	Crop and Cultivar	Density	Sowing Date	Harvest Window
A1 to A2 (Spring)	Sugar snap peas (Sugar Ann)	8 per sq	Mar 15 (sow)	May 20 to Jun 10
A1 to A2 (Summer)	Kentucky Wonder pole beans (after peas)	4 per sq	Jun 10 (sow)	Aug 15 to Oct 10
A3 to A4	Marketmore 76 slicing cucumber	2 per sq	May 20 (transpl.)	Jul 15 to Sep 15
A5 to A7	San Marzano paste tomato	1 per sq	May 10 (transpl.)	Jul 25 to Sep 30
A8	Indeterminate cherry tomato	1 per sq	May 10 (transpl.)	Jul 5 to Oct 15
B1 to B2, C1 to C2	Bush snap/green beans (Provider)	4 per sq	May 20 (sow)	Jul 10 to Aug 15
B3 to B5	Sweet bell pepper (California Wonder)	1 per sq	May 15 (transpl.)	Aug 1 to Oct 15
B6 to B7	Genovese basil	4 per sq	May 20 (transpl.)	Jun 25 to first frost
B8	Curly parsley	4 per sq	Apr 1 (transpl.)	May 15 to Nov 15
C3 to C4	Japanese eggplant	1 per sq	May 20 (transpl.)	Jul 20 to Oct 1
C5 to C6	Swiss chard (Bright Lights / Fordhook)	4 per sq	Apr 15 (sow)	Jun 10 to hard frost
C7	Culinary thyme (perennial)	1 per sq	Apr 15 (transpl.)	Year-round snipping
C8 (Spring / Fall)	Golden Acre cabbage, then Baby spinach	1, then 9/sq	Apr 1 / Sep 1	Jun 20 to Jul 10 / Oct
D1, D8	French marigold (*Tagetes patula*)	1 per sq	May 15 (transpl.)	Bloom until frost
D2 to D4	Danvers half-long carrot (Spring and Fall)	16 per sq	Apr 1 / Aug 1	Jun 15 / Oct 20
D5 to D6	Scallions (slender green bunching onions)	16 per sq	Apr 15 (sow)	Jun 15 onward
D7	Table beets (Detroit Dark Red)	9 per sq	Jun 1 (sow)	Aug 15 to Sep 30

MASTER 32-CELL FIELD PLANNING WORKSHEET (BP-12)

SCALE 1:16 • BLANK PRINTABLE MASTER GRID

N

BED ID: _____ | TARGET SOW DATE: _____ | SOIL TEMP: _____ °F | FIRST HARVEST: _____

	1	2	3	4	5	6	7	8
A					CENTER CROSS-TIE			
B								
C								
D								

Protocol: Water 6–9 AM at soil level | Mulch w/ 2–3 in clean herbicide-free straw in late June (verify no persistent residues)

1:16 PRINTABLE GRID | SHEET BP-12